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Reg. No. | | | | | | | |

Time : 3.00 hrs.

Quarterly Examination - 2025

PHYSICS

Max. Marks : 70

PART - I

I. Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer. **15 x 1 = 15**

1. A parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?
   a) capacitance b) charge c) voltage d) energy density
2. When glass rod rubbing with silk cloth the glass rod acquire the charge is
   a) positive b) negative c) zero d) none of these
3. In Joule's heating law, when R and t are constant, if the H is taken along the y axis and I^2 along the x axis, the graph is
   a) straight line b) parabola c) circle d) ellipse
4. The temperature coefficient of resistance of a wire is 0.00125 per ^{0}C. At 20^{0}C, its resistance is 1Ω. The resistance at the wire will be 2Ω at
   a) 800^{0}C b) 700^{0}C c) 850^{0}C d) 820^{0}C
5. The vertical component of Earth's magnetic field at a place is equal to the horizontal component. What is the value of angle of dip at this place?
   a) 30^0 b) 45^0 c) 60^0 d) 90^0
6. If a material having intensity of magnetisation 500 Am^{-1} is placed in a magnetising field of 1000 Am^{-1} then the susceptibility of the material is
   a) 0.2 b) 0.8 c) 0.7 d) 0.5
7. When the current changes from +2A to –2A in 0.05s, an emf at 8V is induced in a coil. The co-efficient of self induction of the coil is
   a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H
8. A step down transformer reduces the supply voltage from 220V to 11V and increase the current from 6A to 100A. Then its efficiency is
   a) 1.2 b) 0.83 c) 0.12 d) 0.9
9. Which of the following electromagnetic radiation is used for viewing objects through fog
   a) microwave b) gamma rays c) x rays d) infrared
10. Fraunhofer lines are an example of.....spectrum.
    a) line emission b) line absorption c) band emission d) band absorption
11. For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is
    a) 30^0 b) 45^0 c) 60^0 d) 90^0
12. Stars twinkle due to
    a) reflection b) total internal reflection c) refraction d) polarisation
13. A plane glass is placed over a various coloured letters (violet, green, yellow, red). The letter which appears to be raised more is
    a) red b) yellow c) green d) violet
14. Calculate the distance upto which ray optics is a good approximation for likght of wavelength 500nm falls on an aperture of width 0.5 mm
    a) 20 cm b) 25 m c) 25 cm d) 30 cm
15. The dimension of $\frac{1}{\mu_o \varepsilon_o}$ is
    a) $[LT^{-1}]$ b) $[L^2 T^{-2}]$ c) $[L^{-1} T]$ d) $[L^{-2} T^2]$

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PART - II

Note : Answer any six questions. Question No.24 is compulsory. **6 x 2 = 12**

16. Define 'electric flux'.
17. Distinguish between drift velocity and mobility.
18. State ampere's circuital law.
19. Define Q factor.
20. Write the limitations of cyclotron.
21. A parallal plate capacitor has two square plates of side 5cm and separated by a distance of 1mm. Calculate the capacitance of this capacitor.
22. What is displacement current?
23. Why does sky appear blue?
24. Light travels from air into a glass slab of thickness 50cm and refractive index 1.5. What is the speed of light in the glass slab?

PART - III

Note : Answer any six questions. Question No.33 is compulsory. **6 x 3 = 18**

25. Derive an expression for the torque experienced by a dipole due to a uniform electronic field.
26. State Kirchhoff's current rule and voltage rule.
27. Give an account of magnetic Lorentz force.
28. Mention the various energy losses in a transformer.
29. What are Fraunhofer lines? How are they useful in the identification of elements present in the sun?
30. What is optical path? Obtain the equation for optical path.
31. Compute the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^4 NC^{-1} and 2×10^{-4} T, respectively.
32. What is power of a lens? Give its unit.
33. The resistance of a nichrome wire at 20^0C is 10Ω. If its temperature coefficient of resistivity of nichrome is $0.004/^0$C. Find the resistance of the wire at boiling point of water. Comment on the result.

PART - IV

Note : Answer all the questions. **5 x 5 =- 25**

34. a) Derive an expression for electrostatic potential due to an electric dipole. **(OR)**
    b) Obtain the expression for electric field due to an infinitely long charged wire.
35. a) Explain the determination of unknown resistance using meter bridge. **(OR)**
    b) Explain the determination of the internal resistance of cell using voltmeter.
36. a) Discuss the working of cyclotron in detail. **(OR)**
    b) Write down Maxwell equations in integral form.
37. a) Explain the construction and working of transformer, **(OR)**
    b) Obtain lens maker's formula and mention its significance.
38. a) Obtain the equation for bandwidth in Young's doubel slit experiment. **(OR)**
    b) Explain the types of emission spectrum.

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